Facial Recognition Drawer Unlock System: Phase 02

ITAI 2277

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System Overview & Foundations

- **Architectural Core: ESP-WHO**
- **Edge AI Framework (ESP-WHO):** A comprehensive development framework for AI applications. It leverages the **MobileFaceNet** model on the ESP32-S3 to perform face detection and recognition locally.
- **Transfer Learning:** Instead of training a model from scratch, we use "One-Shot Learning." The system is pre-trained to understand "human faces" in general, needing only 5–10 samples to recognize a specific user.

Frameworks, Models, and Datasets

- **The Datasets:** Massive collections like **WIDER FACE** and **VGGFace2** (millions of images). These stay on high-power servers and are never stored on the ESP32.
- **The Models (The Distilled Knowledge):** Small, mathematical files (weights) included in **ESP-WHO**. think of this as the "intelligence" extracted from the datasets—compact enough to fit in the ESP32-S3's memory.
- **The Framework (The Toolbox): ESP-WHO** provides the structure and functions (like align_face() or enroll_face()) that allow you to use these models effectively in your C++ code.

Component	Dataset	Scale	Purpose
Detection (MTMN)	**WIDER FACE**	393k Faces	Enables detection despite masks, glasses, or low light.
Recognition (FR)	**VGGFace2**	3.3M Images	Learns universal facial "deltas" (bone structure, etc).
Alignment	**AFW / PASCAL**	1.8k Faces	Finds landmarks (eyes, nose) to "level" the face.

Models Present in the ESP32-S3

MTMN (Multi-task Cascaded Convolutional Networks)

- The **MTMN (Multi-task Cascaded Convolutional Networks)** model which is used for face detection in ESP-WHO was primarily trained on the **WIDER FACE** dataset.
- **Scale:** 32,203 images containing 393,703 labeled faces.
- **Purpose:** This dataset features high degrees of variability in scale, pose, and occlusion. This is why the ESP32 can detect a face even if you are wearing a mask, glasses, or standing in poor lighting.

MobileFaceNet

- The **MobileFaceNet** architecture, which generates the 512-bytefeature vector (embedding) we are using, was trained on these two massive datasets:
- **MS-Celeb-1M:** Contains 10 million character images of 100k celebrities. It provides the model with a "deep dictionary" of human facial features.
- **VGGFace2:** Contains 3.31 million images of 9,131 subjects. It is specifically designed to handle large variations in pose and age.
- **Role in the project:** These datasets taught the AI what a "human face" looks like generally, so that when it sees our face, it only needs to learn the specific mathematical differences (the "delta") that make us unique.

How It Works: Transfer Learning

- **One-Shot Learning at the Edge**
- The XIAO ESP32-S3 Sense uses **Transfer Learning** to recognize you without needing a supercomputer.
- **Pre-trained Logic:** By including #include "fr_forward.h", you load a "brain" that already understands human features (eyes, nose, jawline) based on its "childhood" training on massive datasets such as **WIDERFACE** and **VGGFace2.**
- **Universal Features:** Because it knows what a *generic* human looks like, it doesn't need to relearn what a face looks like—it only needs to learn your unique measurements.
- **Local Enrollment:** When you save your face, you aren't "training" a new model. You are simply giving an "educated" AI a new mathematical key to remember in its **PSRAM**.

Data Collection & Engineering Pipeline

The system follows a strict pipeline to convert a camera frame into a secure lock trigger:

Capture: Raw JPEG from the OV2640 sensor.

Preprocessing: Decompression into an **RGB888** matrix.

Detection: MTMN identifies the bounding box and 5 facial landmarks.

Alignment: Affine Transformation straightens the face (correcting for head tilt).

Feature Engineering: Conversion into a **512-byte (256-dimension)** mathematical feature vector.

Testing Data Acquisition

- We utilized the Labeled Faces in the Wild (LFW) dataset to establish a robust security baseline.
- The system was tested against 3,023 samples representing 62 unique individuals.
- While team photos will be enrolled via the mobile framework, the LFW set serves as our "Unauthorized" population to test rejection reliability.

Technical Implementation (Data Collection)

- **C++ Logic for Face Enrollment**
- This routine runs on the **XIAO ESP32-S3 Sense** to populate the local PSRAM database.

```
// 1. Feature Engineering: Alignment & Embedding Generation
// This generates the 512-byte (256 * 2 bytes) feature vector
dl_matrix3du_t *aligned_face = dl_matrix3du_alloc(1, 112, 112, 3);

if (align_face(net_boxes, image_matrix, aligned_face) == ESP_OK) {
    // 2. Enrollment (Data Storage in PSRAM)
    // The model measures Euclidean Distance for future matches
    int8_t left_sample = enroll_face(&id_list, aligned_face);
    Serial.printf("Sample Collected. Samples remaining: %d\n", left_sample);
}
```

Privacy, Ethics, and Validation

Secure Biometric Standards

- **Local-Only Storage:** No images or personal data ever leave the device.
- **Data Minimization:** We store **vectors**, not photos. It is mathematically impossible to reconstruct a face from the 512-byte hash.

- **Validation Milestone:**
  - **Splitting:** We will use the "Leave-One-Out" cross-validation during testing. Collect 5 samples; test if 4 samples can correctly recognize the 5th.
  - **Goal:** Achieve a **False Acceptance Rate (FAR) < 1%**.
  - **Final Output:** A `face_id_list` binary blob saved to Flash memory.

Data Preprocessing

- **Technical Goal:** Optimizing raw image data for the ESP32-S3's limited computational resources.
- **Standardization:** Images were resized to a 0.4 ratio, resulting in 1,850 features (pixels) per image.
- **Detection & Alignment:** We utilized a pipeline to detect facial presence and normalize features, which is critical for consistent mathematical encoding.

Exploratory Data Analysis

- **Security Metric:** We calculated Euclidean Distances between facial encodings to define the boundary between an "Authorized" match and a "Stranger".
- **Findings:** Our analysis shows that unauthorized individuals (strangers) have a distance score distribution primarily between 8 and 16.
- **Unlock Threshold:** We have established a strict Unlock Threshold of 20 to ensure high security and minimize false acceptance.